STATSEI14 · PRACTICAL TUTORIAL

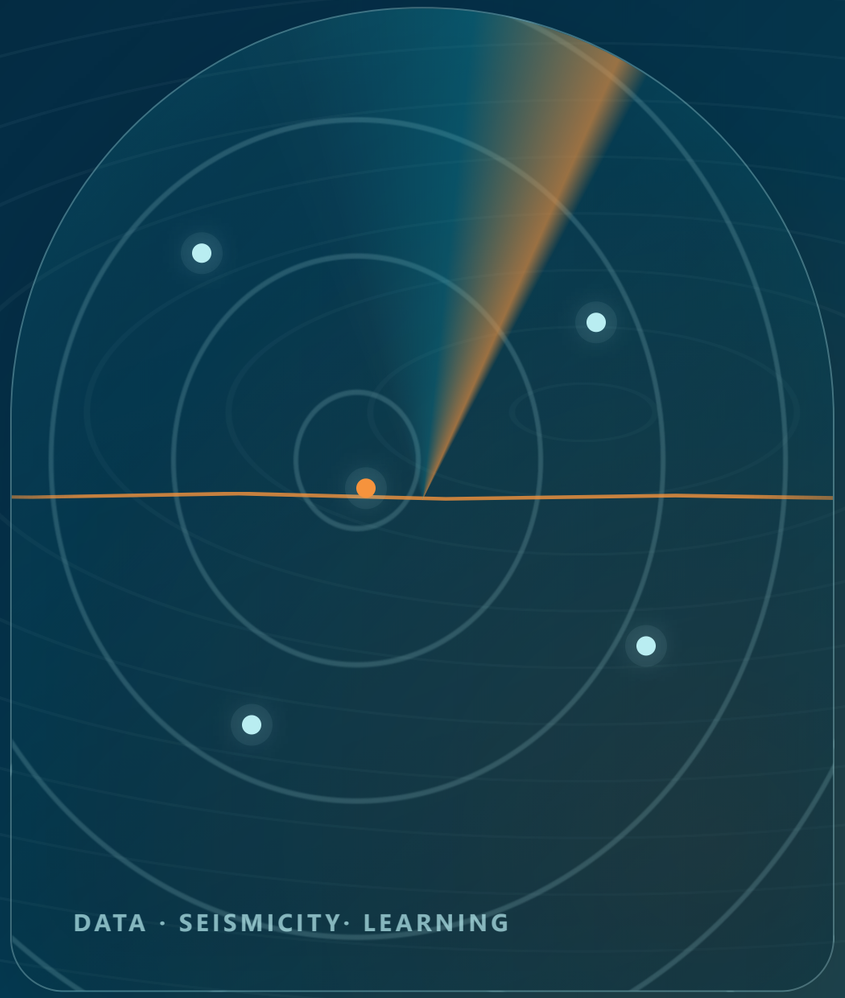
# Deep Learning Meets ETAS

## Practical Hybrid Workflows for Seismic Forecasting

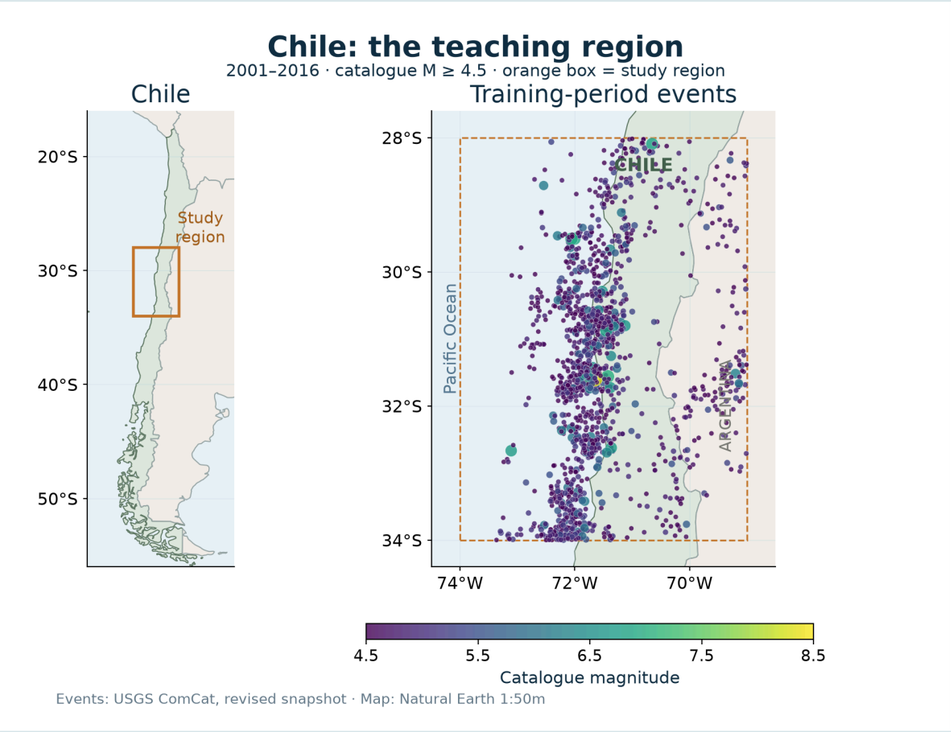
**Francisco Plaza-Vega**
Universidad de Santiago de Chile

13 October 2026 · Chile

> Seismic learning tasks · Research cases · Guided modelling and discussion



# What can we learn from seismicity?




**Signals** reveal recorded events.

**Catalogues** describe event histories.

**Models** turn histories into questions about what may follow.

> What information do we represent, and what do we ask a model to learn?

Map: frozen USGS training catalogue and Natural Earth. Illustrates today's regional setting.

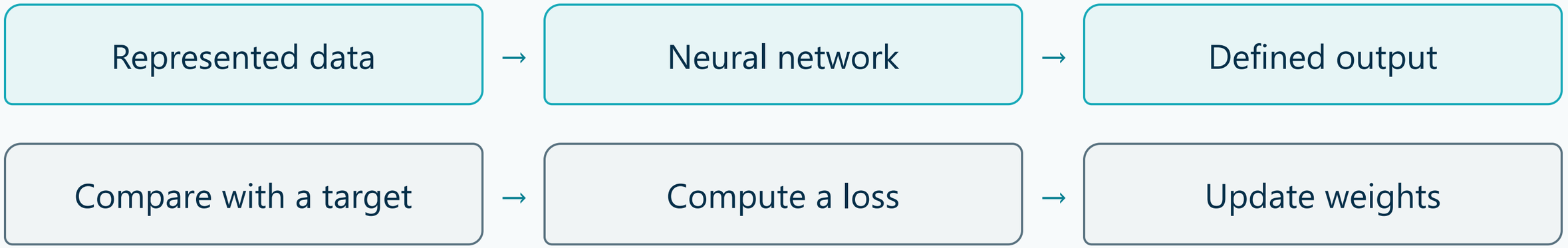
# Deep learning in seismology: inputs and targets

| Example | Input | Learning task |
|---|---|---|
| **ConvNetQuake · 2018** | Recorded waveforms | Detect events and classify their source region |
| **RECAST · 2023** | Past event times and magnitudes | Estimate the next-event time distribution |
| **Delgado et al. · 2025** | Features of catalogued events | Classify foreshock, mainshock and aftershock labels |
| **Farfán et al. · 2025** | Sequences of daily spatial grids | Forecast gridded seismic activity |

> Today’s practical: 30 days of history and a seven-day occurrence probability.

Perol et al. · Dascher-Cousineau et al. · Delgado et al. · Farfán et al.

# A network learns a mapping from examples

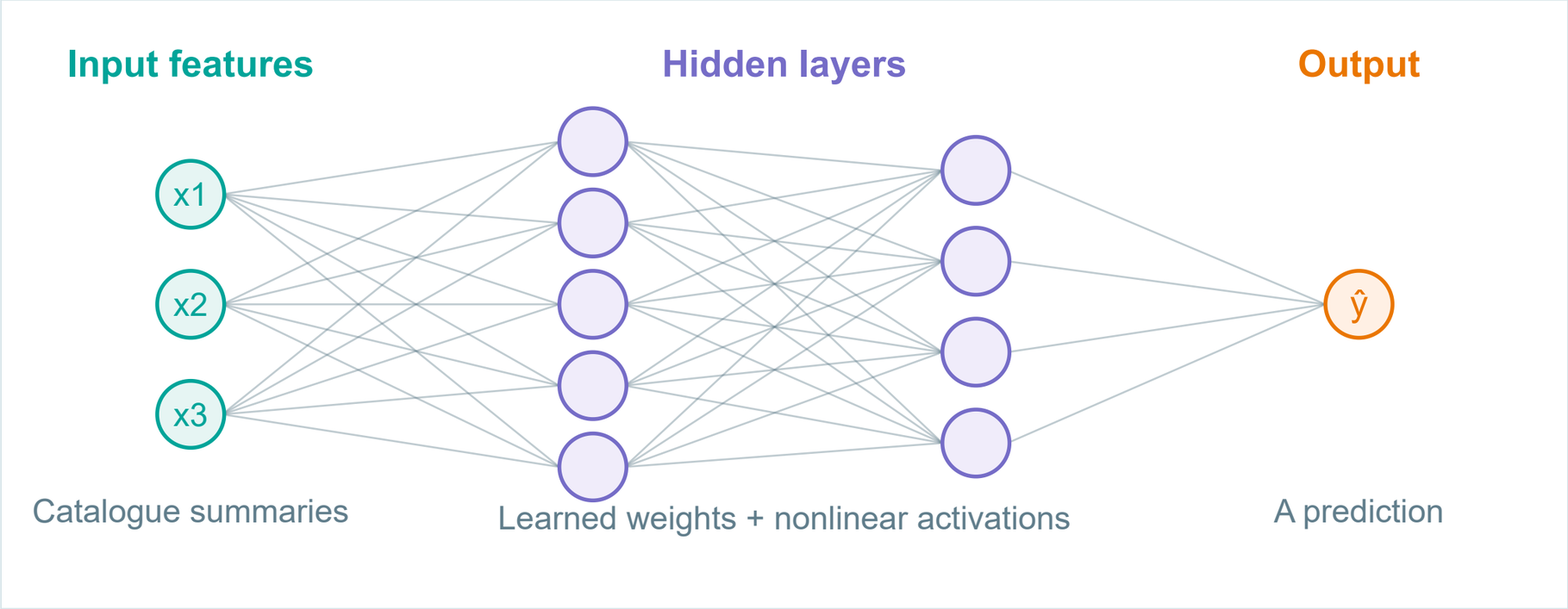



**Architecture** determines how information is combined.

**Training** adjusts weights to reduce the loss.

> The output and the loss must match the scientific question.

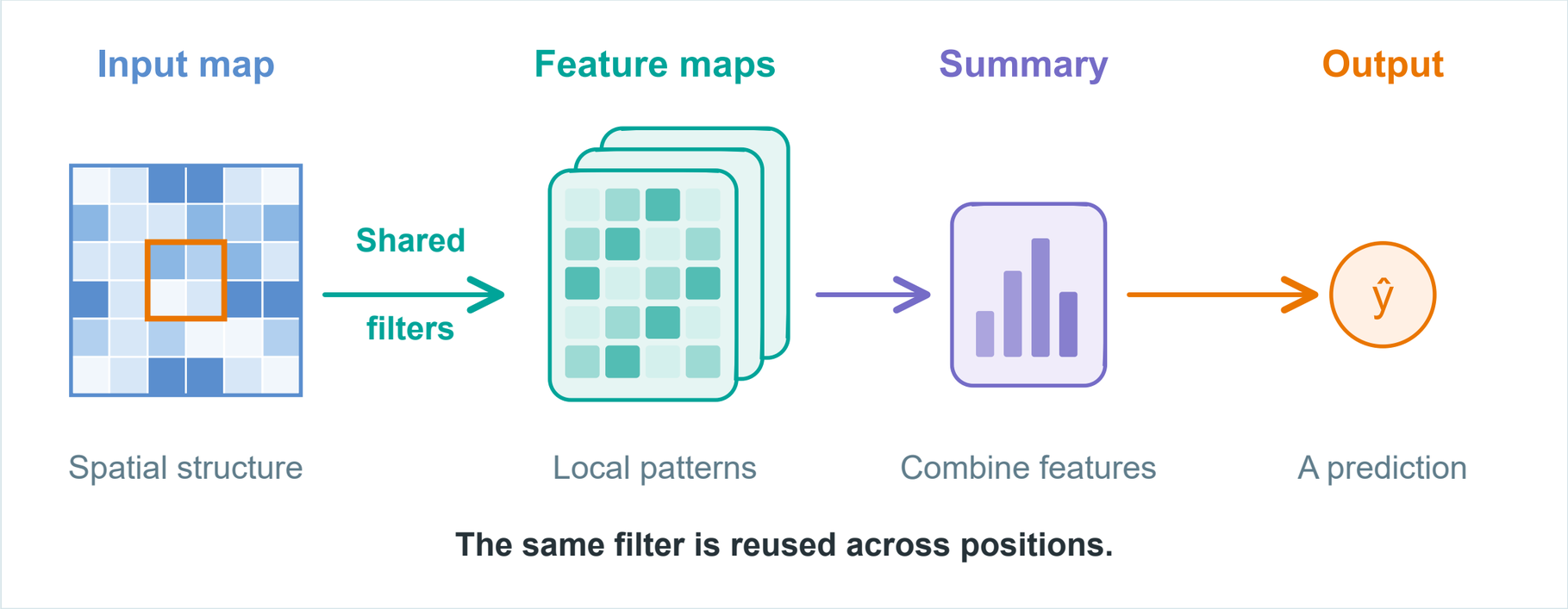
# MLP: combine measured features




A multilayer perceptron learns nonlinear combinations of a feature vector.

Optional extension: the MLP uses 90 values at fixed lag positions. Diagram adapted from the author's DL course, Unit 2.

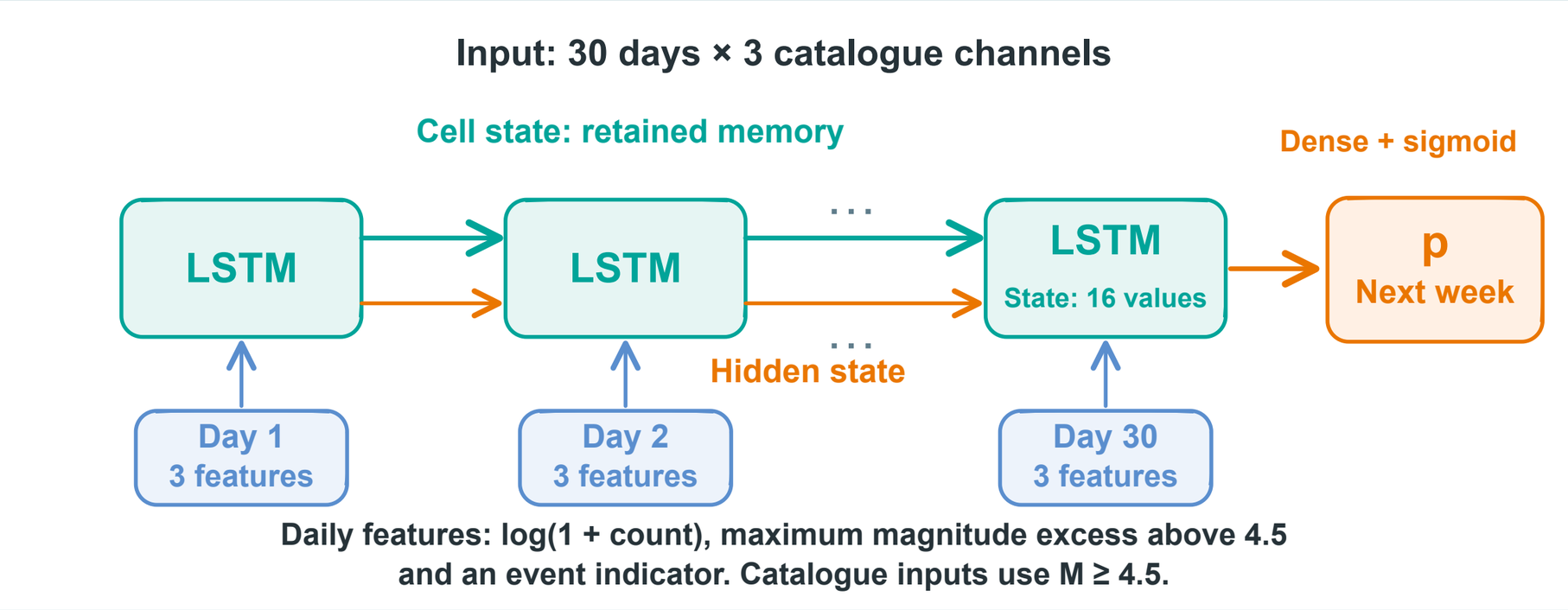
# CNN: learn local patterns with shared filters




The same filter examines different neighborhoods.

Optional extension: CNN1D shares filters across days. Diagram adapted from the author's DL course, Unit 3.

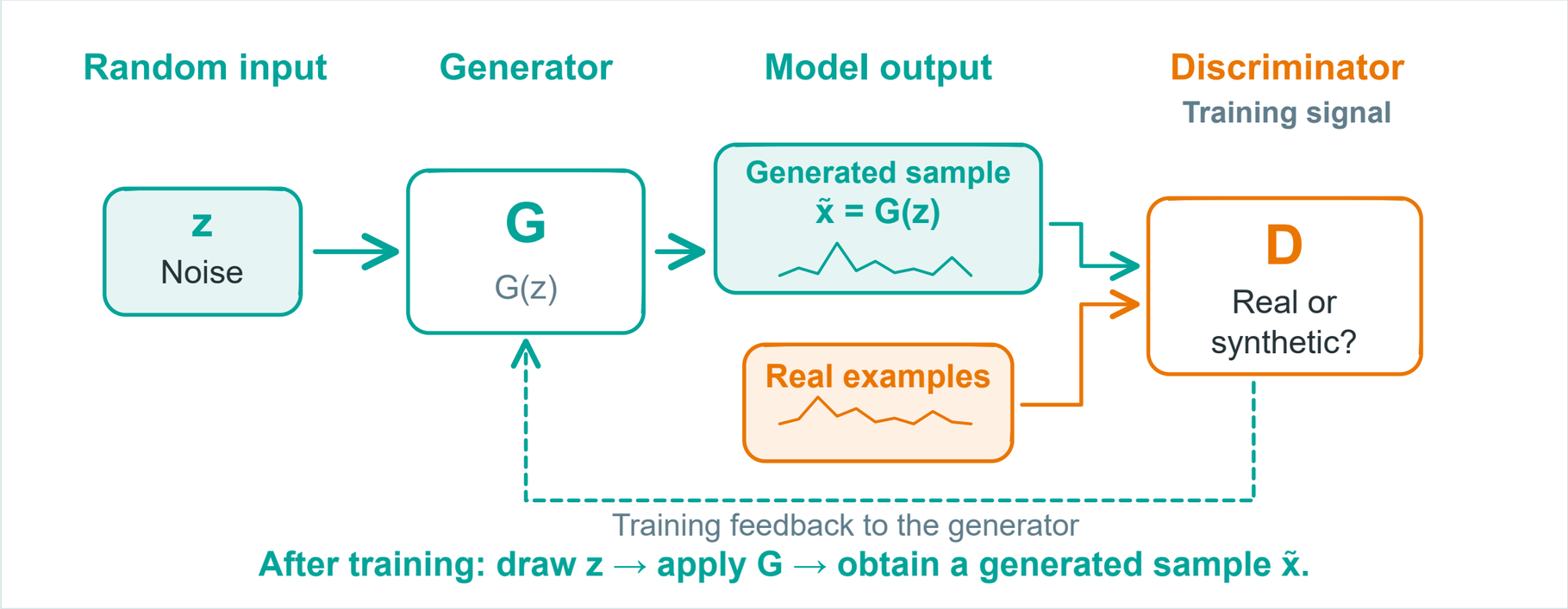
# LSTM: carry information through a sequence




Today's example: 30 × 3 input → state (16) → Dense + sigmoid → weekly probability

Output: probability of at least one catalogue M ≥ 5 event in the next seven days.
Adapted from the author's DL course, Unit 4.

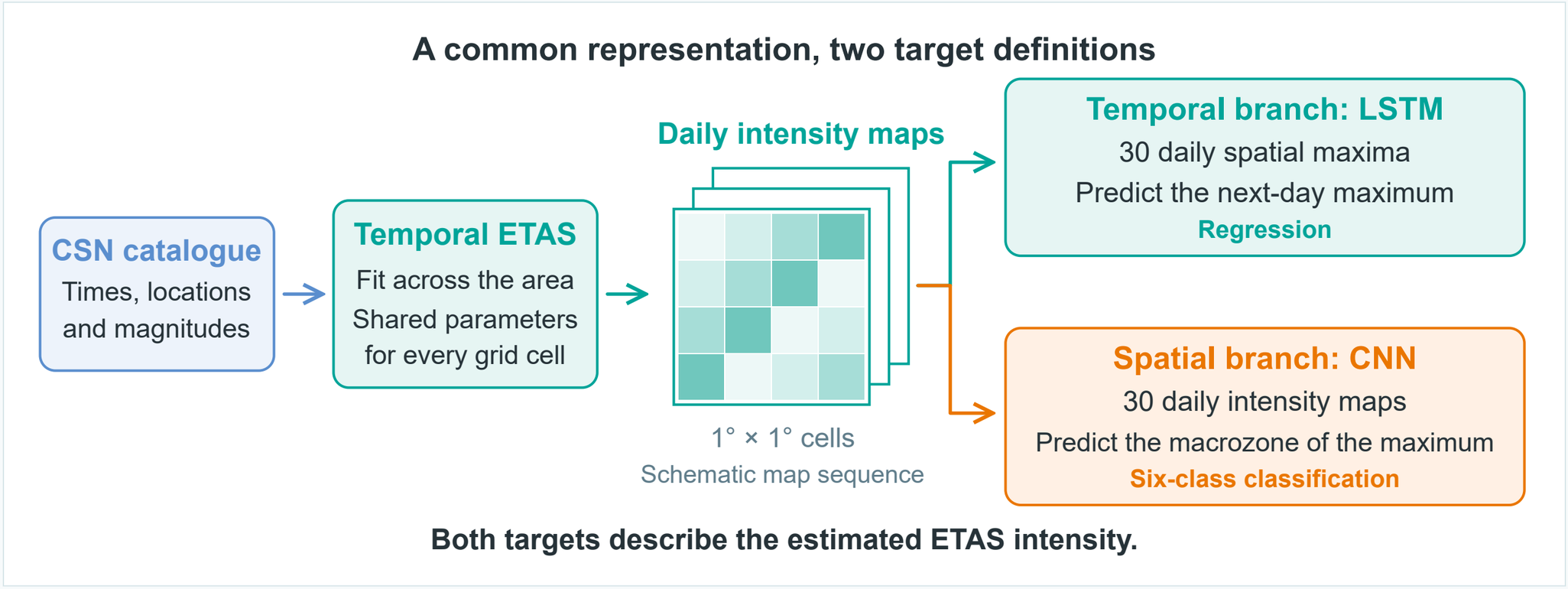
# Generative learning: produce new samples




> The output of G is a new data sample, such as a synthetic sequence.

Later case: mainshock conditions + noise → a possible aftershock sequence.
GAN schematic adapted from the author's DL course, Unit 5.

# ETAS intensity maps support two learning tasks




Nicolis, Plaza & Salas · Spatial Statistics 42 (2021), 100442 · CSN Chile, 2000–2017

# ETAS intensity describes a conditional event rate

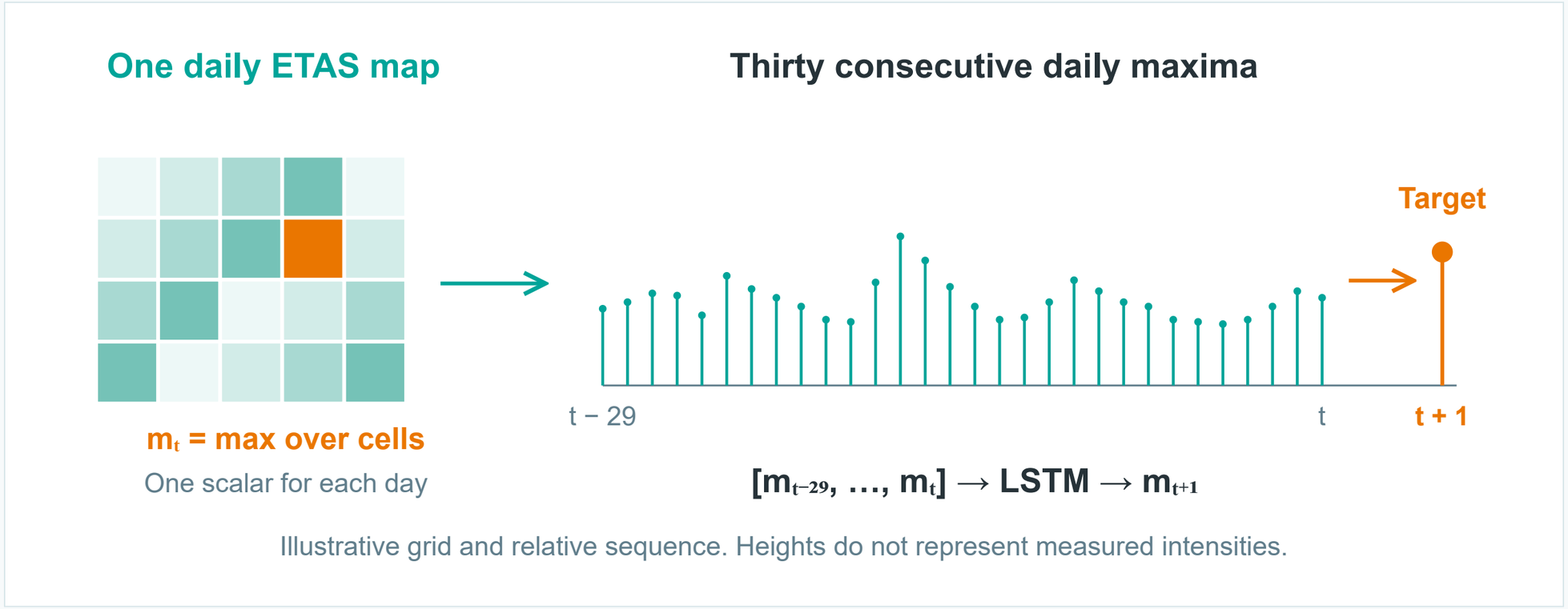
$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \text{contribution from event } i$$

Background + magnitude-dependent contributions that decay with time

**Published construction:** common ETAS parameters applied across 1° × 1° cells produce a sequence of intensity maps.

> A map value is an estimated occurrence rate, not an earthquake magnitude.

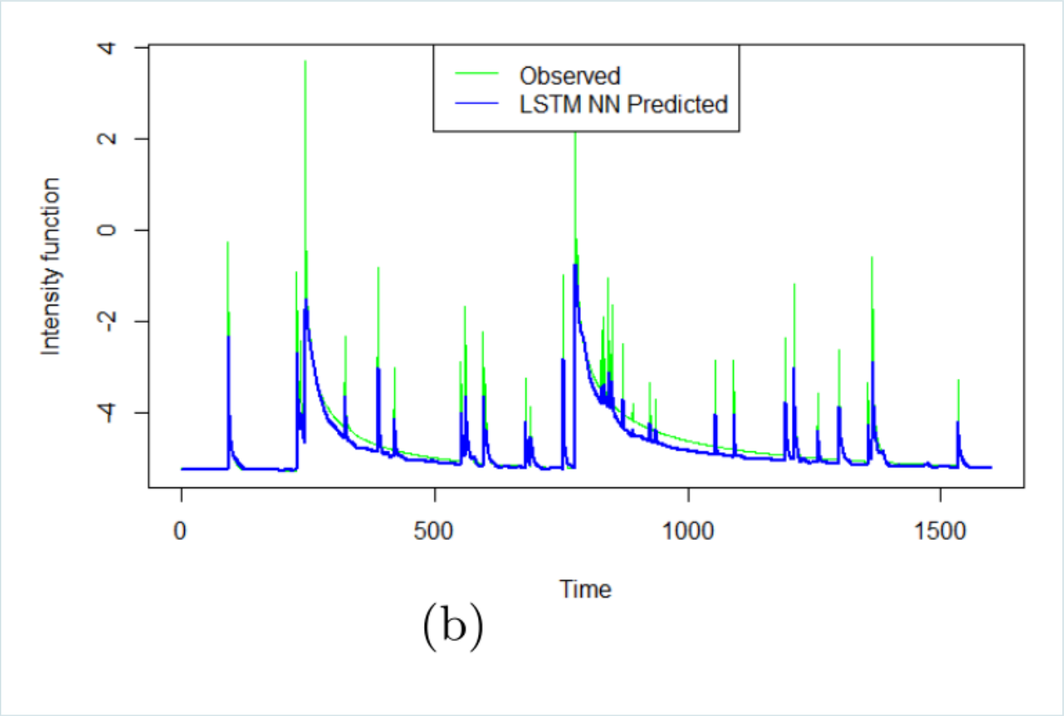
# A map maximum becomes one value in the LSTM input




30 daily spatial maxima → LSTM → next-day maximum intensity

Nicolis, Plaza & Salas (2021) · Original published regression task · Diagram illustrates the operation, not measured values

# Published LSTM result: an ETAS-derived test target




## M ≥ 6 setting · Table 3

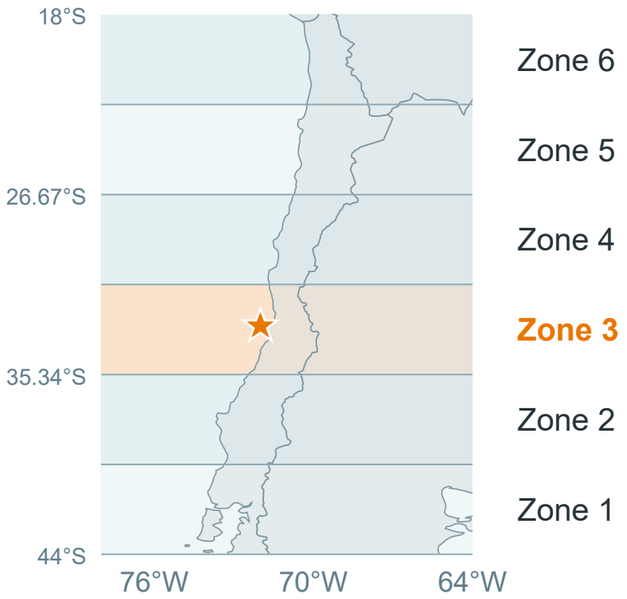
| Model | RMSE | $R^2$ |
|---|---|---|
| ETAS | 0.5475 | 0.6241 |
| LSTM | 0.5296 | 0.6561 |

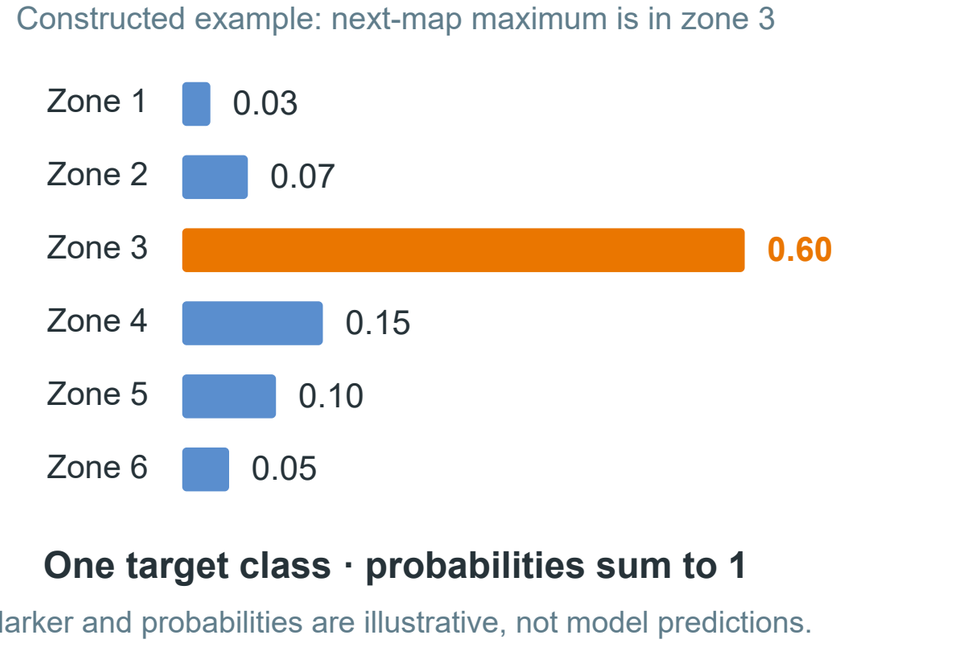
Lower error and higher $R^2$ for this reported comparison.

> The network tracks a model-derived target; this is not a direct event-probability test.

Nicolis, Plaza & Salas (2021), Figure 5(b) and Table 3 · Original plotting scale retained

# CNN: locate the next map maximum in one of six macrozones




The target is one macrozone containing the next day's maximum ETAS intensity.

Nicolis, Plaza & Salas (2021), Table 2 · Natural Earth geography · Marker and probabilities are illustrative

# Generative exploration of aftershock sequences

**Mainshock conditions**

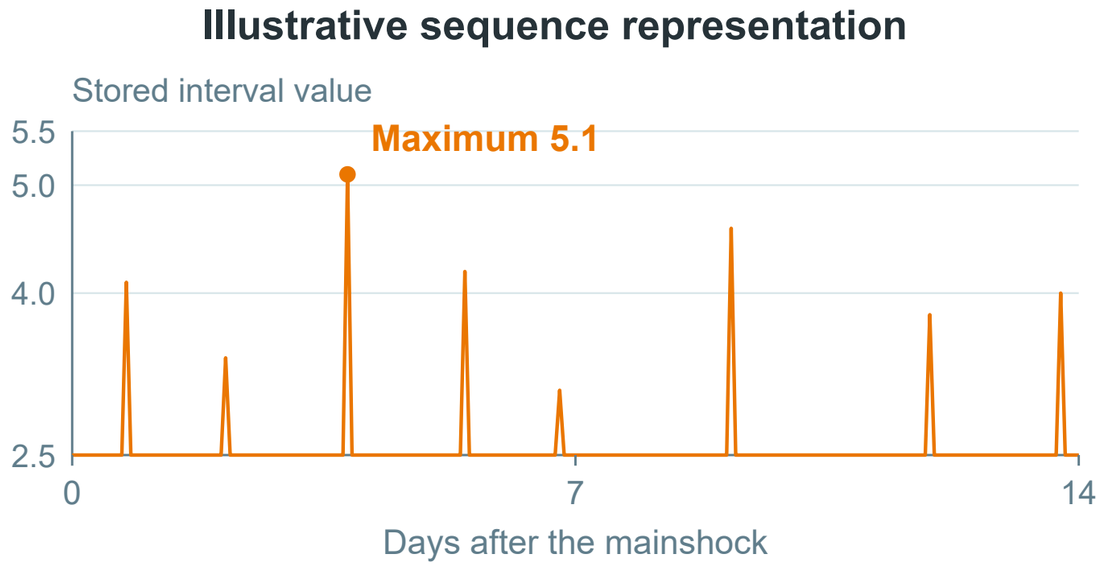
Location, magnitude, depth

**+ random noise**

↓

**Conditional generator** →

cGAN / WGAN-GP / TGAN




**224 intervals of 90 minutes. The value 2.5 marks absence.**

Values chosen for teaching; no empirical forecast is shown.

> The intended output is a possible sequence conditioned on a mainshock.

Jara Muñoz (2025) · Undergraduate thesis in Statistical Engineering · Universidad de Santiago de Chile

# An observed and generated sequence: one manuscript case

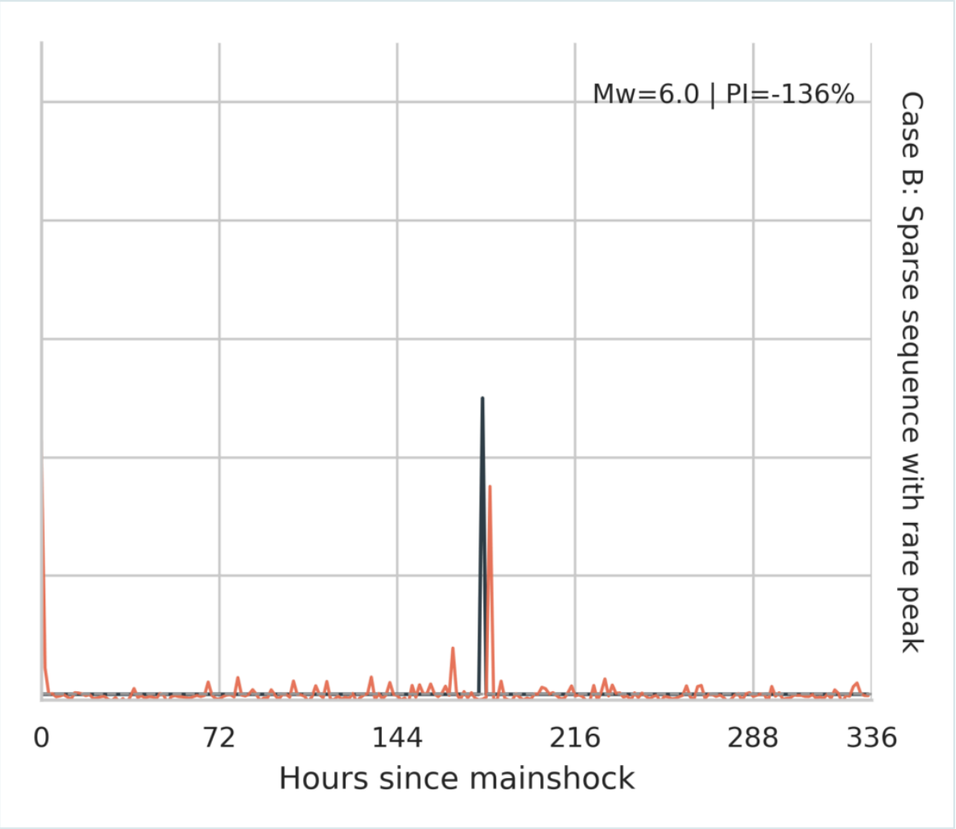
**GANT · case B**

—— Observed reference

—— Generated sequence

Vertical axis: interval maximum magnitude (Mw).

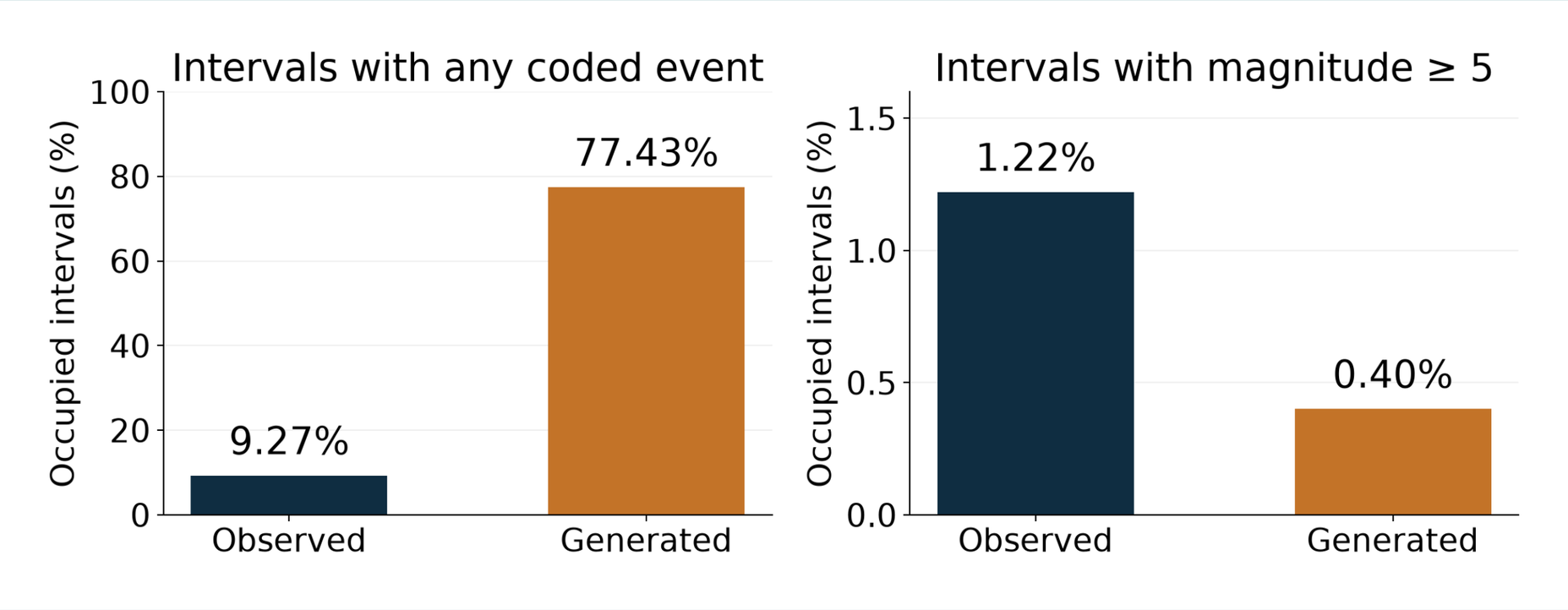
**2.5 marks absence.**




> Compare occurrence, peak magnitude and timing, as well as the shape of the curve.

2026 manuscript under review, Figure 4 · One case from the primary random split · Original PI = −136% for this case

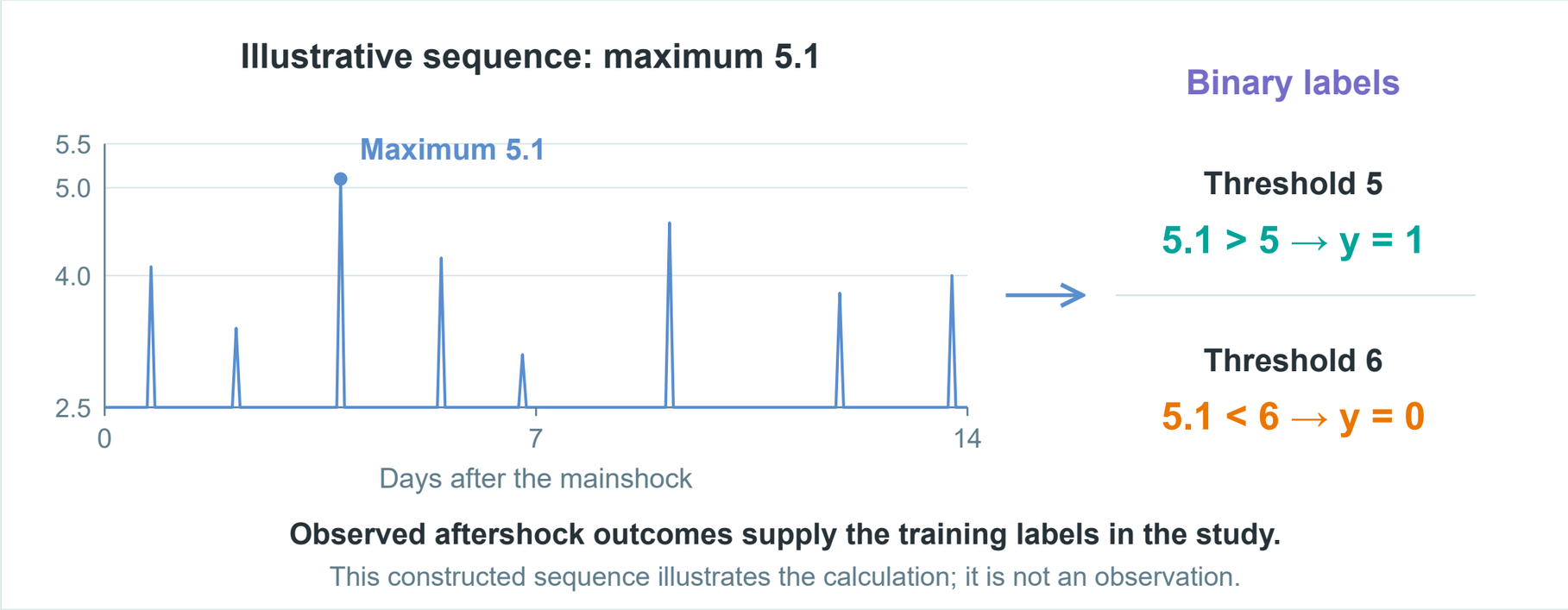
# GANT temporal evaluation: too many occupied intervals




GANT overproduces occupied intervals while underrepresenting intervals with M ≥ 5.

2026 manuscript under review · Temporal evaluation · Percentages of intervals, not total event counts

# Observed sequences define magnitude-threshold labels




**Observed aftershock outcomes supply the training labels in the study.**

This constructed sequence illustrates the calculation; it is not an observation.

> Mainshock features → classifier → probability of at least one qualifying aftershock

Manuscript under review · 14 days, 100 km · Observed outcomes define labels; the displayed sequence is constructed

# Classification performance depends on the evaluation

**Reported target (Table 9):** at least one Mw ≥ 6 aftershock within 14 days and 100 km.

| Evaluation scheme | Best reported classifier | ROC-AUC |
|---|---|---|
| Random split | Random Forest | 0.816 |
| Temporal split | Gradient Boosting | 0.672 |
| Geographic split | Random Forest | 0.642 |

**Each row selects the best model in its own evaluation.** Models and evaluation samples differ.

> These AUCs measure discrimination. They do not establish calibrated probabilities.

Plaza-Vega, Jara, Nicolis & Salinas (2026), manuscript under review · Table 9 · Magnitude-threshold classification

# Research hypothesis: ETAS-informed generation

History available at the origin → ETAS-derived information → Conditional generator

**Possible role:** supply a conditioning variable or a constraint on generated sequences.

**Question:** does that addition improve occurrence, magnitude and temporal structure under a fixed evaluation?

> This is a proposed research direction. No hybrid results are shown here.

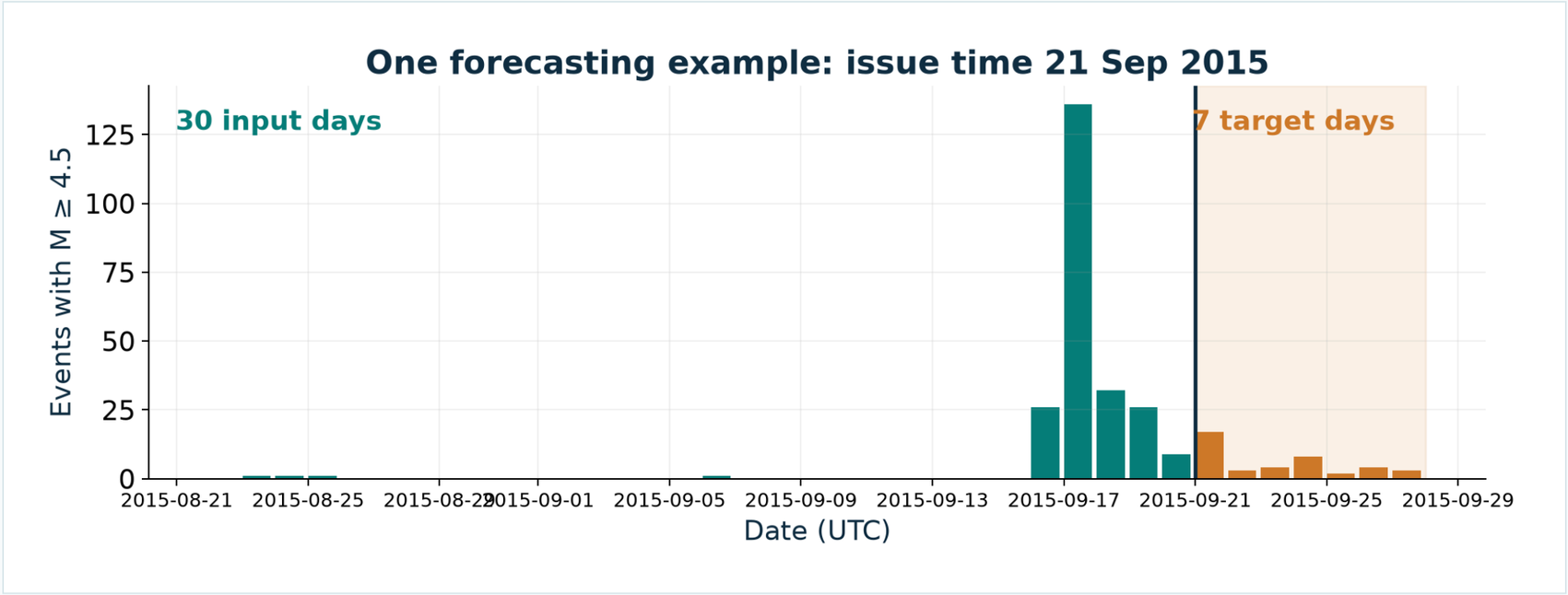
Future work discussed in the manuscript · The tutorial's illustrative activity score is a separate feature experiment

# The practical applies the same modelling decisions

| Case | Representation | Output |
|---|---|---|
| **Spatial Statistics** | ETAS maps and daily maxima | Intensity maximum and its macrozone |
| **Generative study** | Mainshock conditions and noise | A possible binned aftershock sequence |
| **Classification study** | Mainshock predictors; observed outcome labels | Occurrence or exceedance probability |
| **Today's practical** | Daily catalogue channels | Weekly occurrence probability |

Core example: references, LSTM and probability evaluation. Optional comparisons: Transformer and spatial models.

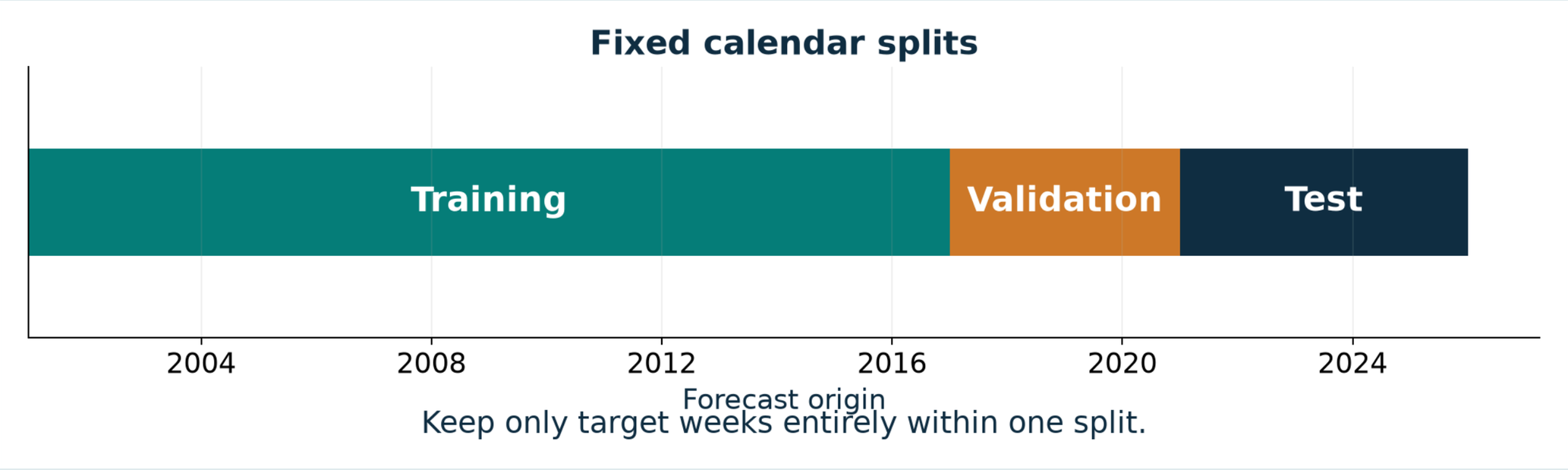
# Today’s target: at least one event next week




30 past days → probability of at least one catalogue M ≥ 5 event in seven days.

One input: 30 days × 3 channels · Region: 28–34°S, 69–74°W · Input M ≥ 4.5 · Monday origins

# Evaluate using the temporal order of the problem




Today we compare validation probabilities using Brier score, log loss and calibration.

Train 2001–2016 · Validation 2017–2020 · Test 2021–2025 was inspected previously and is excluded from this comparison.

# Guided experiment: from one example to an LSTM probability

## 30 days × 3 catalogue channels → LSTM → weekly probability

**Build one example:** identify the input window, forecast origin and label.

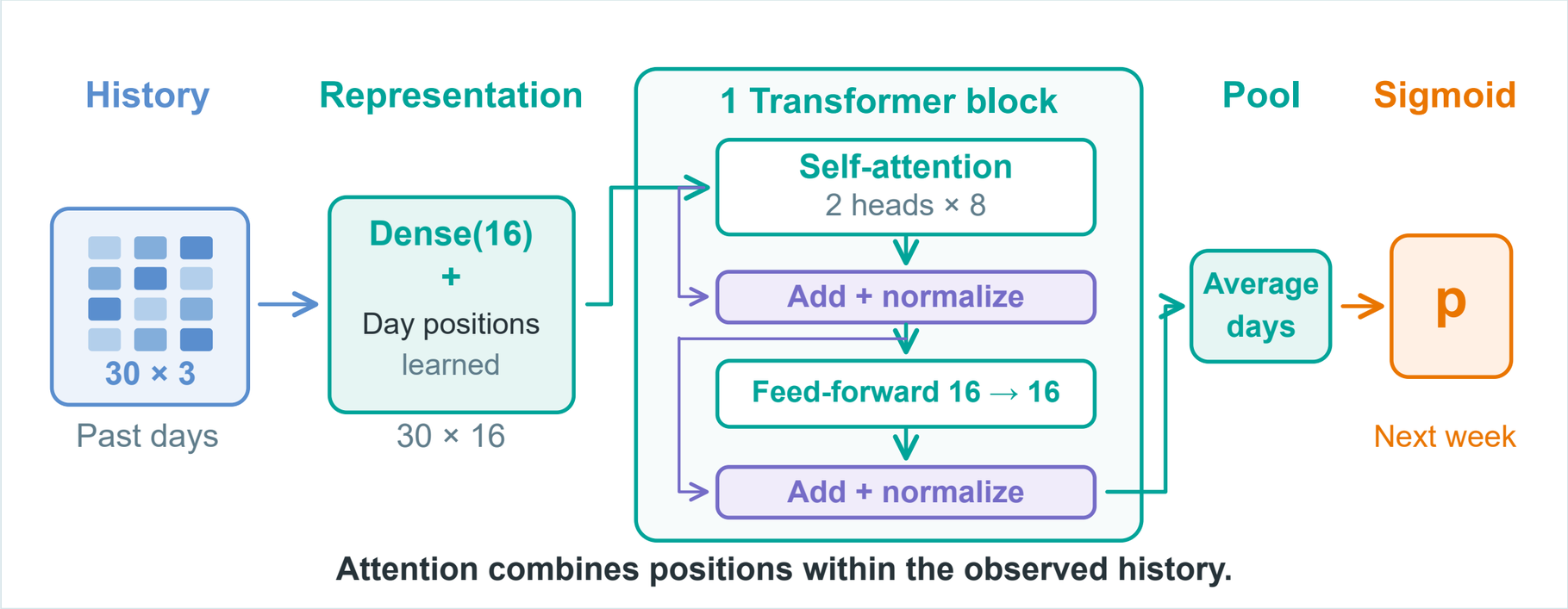
**Fit references and the LSTM:** read compile, fit and predict beside the model.

**Evaluate:** compare Brier score, log loss and calibration.

> Does the LSTM improve on training prevalence and logistic regression in this experiment?

Temporal tutorial and notebook · Seed 7 · Optional comparison: Transformer on the same history and target

# Optional comparison: a Transformer on the same history




Same history and target. A different way to combine information across days.

Current temporal tutorial · 2,257 trainable parameters versus 1,297 for LSTM · One fixed, small configuration

# Validation performance for the weekly occurrence task

| Model | Validation Brier | Validation log loss |
|---|---|---|
| Training prevalence | 0.188 | 0.563 |
| Logistic regression | 0.204 | 0.614 |
| LSTM | 0.182 | 0.548 |
| Transformer (optional) | 0.188 | 0.562 |

**Lower is better.** The LSTM has the lowest losses in these individual fits. Read calibration alongside them.

> These are development results from one region and one validation period.

One fit per neural model, seed 7 · 208 validation weeks · Display rounded to three decimals

# What can this result tell us about the problem?

## Reading the validation results

Compare each model with training prevalence.

Read Brier, log loss and calibration together.

## Questions for the next experiment

Which representation and target preserve useful structure?

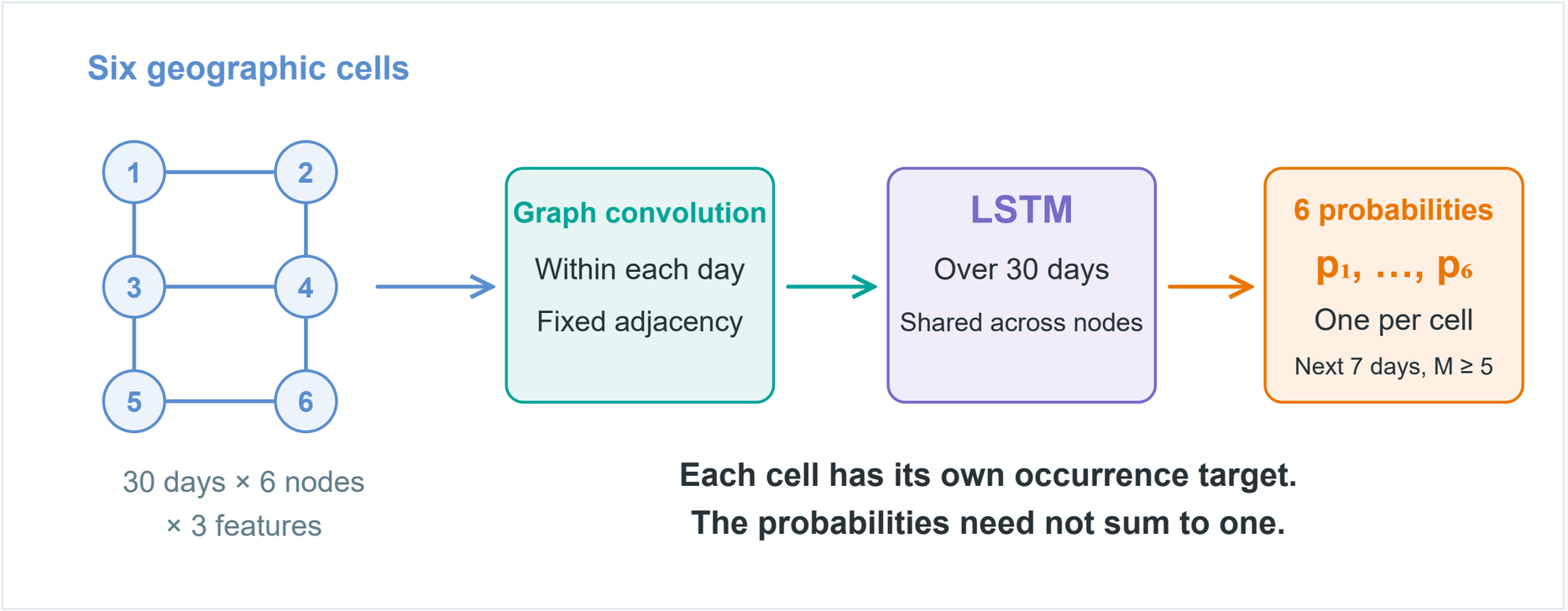
What information is available in the catalogue?

Would more data, capacity or a different training budget change the result?

> Useful deep learning depends on how we represent the phenomenon and formulate the task.

Validation also controls early stopping. One seed does not measure initialization sensitivity.

# Spatial extension: Graph + LSTM




Graph connections share spatial information; the LSTM summarizes each node's history.

Input/output introduction · Complete graph example for optional exploration

# What did we learn, and what would we test next?

1. What information did our daily channels preserve or discard?
2. What did the LSTM comparison with simple references establish?
3. Which research question would you explore next with maps, graphs or an activity score?

> Deep learning offers flexible tools. Their scientific value depends on the question and the evidence.

Further exploration: optional reading and experiments · LSTM references and optional extensions

# Appendix · the LSTM steps

```
inputs = keras.Input(shape=(30, 3))
state = keras.layers.LSTM(16)(inputs)
probability = keras.layers.Dense(1, activation="sigmoid")(state)
model_lstm = keras.Model(inputs, probability)
model_lstm.compile(optimizer="adam", loss="binary_crossentropy")
history_lstm = model_lstm.fit(
    X_train, y_train, validation_data=(X_val, y_val),
    epochs=20, batch_size=64, callbacks=[stopping_lstm], shuffle=False)
p_lstm = model_lstm.predict(X_val).ravel()
brier = np.mean((p_lstm - y_val) ** 2)
```

> Define the model · compile · fit · predict · evaluate probabilities

# Appendix · quantities and scientific meaning

| Quantity | Question answered |
|---|---|
| **Process intensity** | At what conditional rate do events occur? |
| **Magnitude** | How large is an event? |
| **Shaking intensity** | What motion or effects occur at a location? |
| **Risk** | What consequences may occur given exposure and vulnerability? |

> Today’s output is a probability of a defined occurrence event.

# Appendix · ETAS-inspired feature experiment

$$r_t(s) = \sum_{t-L \le t_i < s} 10^{\alpha(M_i - M_0)} \left(1 + \frac{s - t_i}{c}\right)^{-p}, \quad s \le t.$$

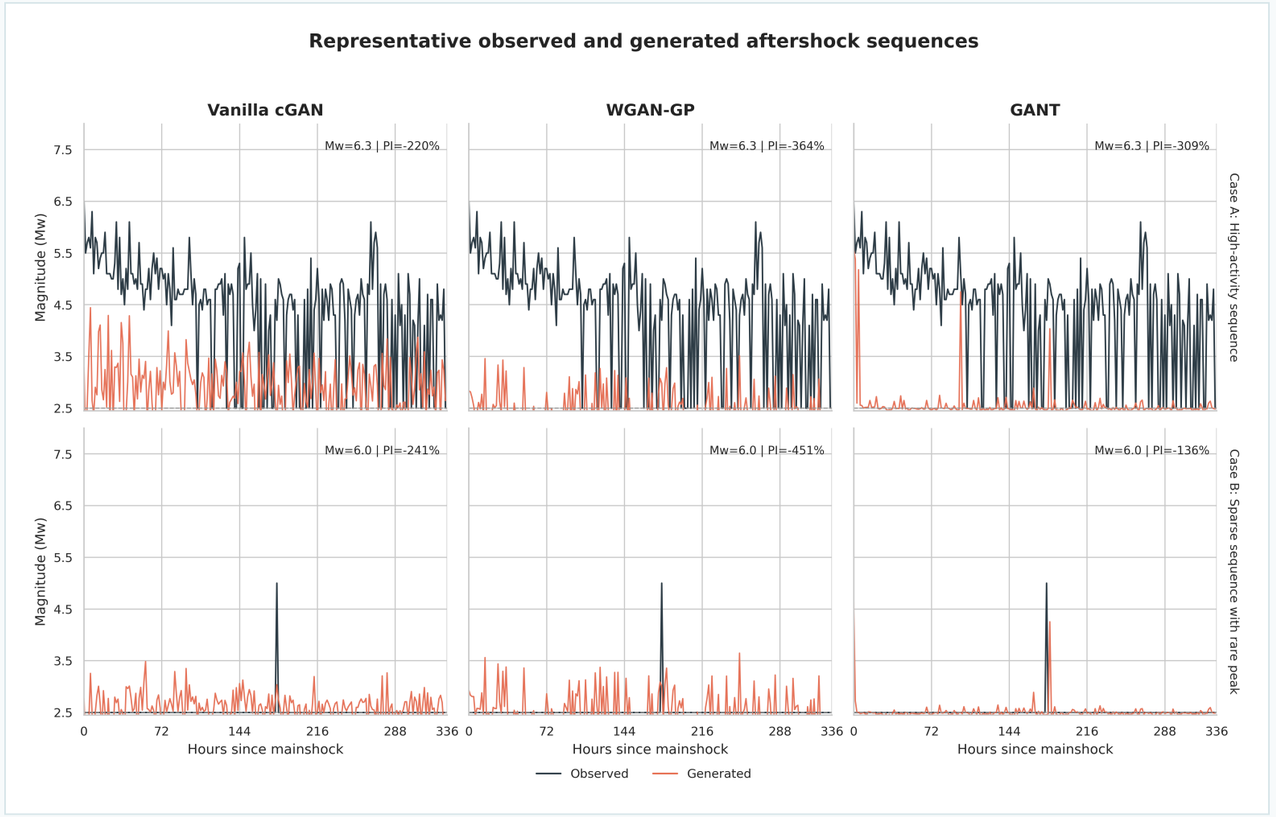
**Illustrative:** α=1, c=1 day, p=1.1, L=30 days.

**Controlled comparison:** both LSTMs use the same available history.

**Representation:** log1p(score) is an additional daily channel.

> Optional representation experiment. The illustrative index is not a fitted ETAS rate.

# Appendix · original generative examples




2026 manuscript under review · Original PI annotations use a mean-reference denominator · Illustrative cases

# Appendix · evidence and availability

| Check | Meaning |
| --- | --- |
| Chronological, purged targets | Future target intervals do not cross partitions. |
| Training-only normalization | Later inputs do not set scaling parameters. |
| Frozen revised catalogue | Data bytes and provenance are reproducible. |
| Historical catalogue vintages | **Not reconstructed.** |

> Reproducibility and historical information availability are separate requirements.

# Appendix · sources and further reading

- **Seismic applications:** ConvNetQuake · RECAST.
- **Our published case:** Nicolis, Plaza & Salas (2021).
- **Aftershock studies:** Jara Muñoz (2025), Statistical Engineering undergraduate thesis, Universidad de Santiago de Chile; Plaza-Vega, Jara, Nicolis & Salinas (2026), manuscript under review.
- **Attention context and further reading:** Delgado et al. (2025) · Farfán et al. (2025).
- **Diagram style:** author's DL course, Units 2–5 · Transformer schematic matches the current temporal tutorial.
- **Materials:** practical · Further exploration · full references.